

4.1 Acceleration and Newton's Second Law

Key Points in this Section.

Differentiation Rules Let $\mathbf{b}(t)$ and $\mathbf{c}(t)$ be differentiable paths in $\mathbb{R}^3$ and $p(t)$ and $q(t)$ be differentiable scalar functions:

$$\text{Sum Rule: } \frac{d}{dt}[\mathbf{b}(t) + \mathbf{c}(t)] = \mathbf{b}'(t) + \mathbf{c}'(t)$$

$$\text{Scalar Multiplication Rule: } \frac{d}{dt}[p(t)\mathbf{c}(t)] = p'(t)\mathbf{c}(t) + p(t)\mathbf{c}'(t)$$

$$\text{Dot Product Rule: } \frac{d}{dt}[\mathbf{b}(t) \cdot \mathbf{c}(t)] = \mathbf{b}'(t) \cdot \mathbf{c}(t) + \mathbf{b}(t) \cdot \mathbf{c}'(t)$$

$$\text{Cross Product Rule: } \frac{d}{dt}[\mathbf{b}(t) \times \mathbf{c}(t)] = \mathbf{b}'(t) \times \mathbf{c}(t) + \mathbf{b}(t) \times \mathbf{c}'(t)$$

$$\text{Chain Rule: } \frac{d}{dt}[\mathbf{c}(q(t))] = q'(t)\mathbf{c}'(q(t)).$$

4.2 Arc Length

Key Points in this Section.

Arc-Length Differential An *infinitesimal displacement* of a particle following a path $\mathbf{c}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ is

$$d\mathbf{s} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k} = \left(\frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k} \right) dt,$$

and its length

$$ds = \sqrt{dx^2 + dy^2 + dz^2} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

is the *differential of arc length*. See Figure 4.2.3.

Arc Length in $\mathbb{R}^n$ Let $\mathbf{c}: [t_0, t_1] \rightarrow \mathbb{R}^n$ be a piecewise C^1 path. Its **length** is defined to be

$$L(\mathbf{c}) = \int_{t_0}^{t_1} \|\mathbf{c}'(t)\| dt.$$

The integrand is the square root of the sum of the squares of the coordinate functions of $\mathbf{c}'(t)$: If

$$\mathbf{c}(t) = (x_1(t), x_2(t), \dots, x_n(t)),$$

then

$$L(\mathbf{c}) = \int_{t_0}^{t_1} \sqrt{(x_1'(t))^2 + (x_2'(t))^2 + \dots + (x_n'(t))^2} dt.$$

4.3 Vector Fields

Key Points in this Section.

Vector Fields A **vector field** in $\mathbb{R}^n$ is a map $\mathbf{F}: A \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ that assigns to each point $\mathbf{x}$ in its domain A a vector $\mathbf{F}(\mathbf{x})$. If $n = 2$, $\mathbf{F}$ is called a **vector field in the plane**, and if $n = 3$, $\mathbf{F}$ is a **vector field in space**.

Flow Lines If $\mathbf{F}$ is a vector field, a **flow line** for $\mathbf{F}$ is a path $\mathbf{c}(t)$ such that

$$\mathbf{c}'(t) = \mathbf{F}(\mathbf{c}(t)).$$

That is, $\mathbf{F}$ yields the velocity field of the path $\mathbf{c}(t)$.

4.4 Divergence and Curl

Key Points in this Section.

1. The **del operator** is

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}.$$

Divergence If $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$, the **divergence** of $\mathbf{F}$ is the scalar field

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}.$$

Curl of a Vector Field If $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$, the *curl* of $\mathbf{F}$ is the *vector field*

$$\begin{aligned}\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} \\ &= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \mathbf{k}.\end{aligned}$$

Basic Identities of Vector Analysis

1. $\nabla(f + g) = \nabla f + \nabla g$
2. $\nabla(cf) = c\nabla f$, for a constant c
3. $\nabla(fg) = f\nabla g + g\nabla f$
4. $\nabla(f/g) = (g\nabla f - f\nabla g)/g^2$, at points $\mathbf{x}$ where $g(\mathbf{x}) \neq 0$
5. $\operatorname{div}(\mathbf{F} + \mathbf{G}) = \operatorname{div} \mathbf{F} + \operatorname{div} \mathbf{G}$
6. $\operatorname{curl}(\mathbf{F} + \mathbf{G}) = \operatorname{curl} \mathbf{F} + \operatorname{curl} \mathbf{G}$
7. $\operatorname{div}(f\mathbf{F}) = f\operatorname{div} \mathbf{F} + \mathbf{F} \cdot \nabla f$
8. $\operatorname{div}(\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot \operatorname{curl} \mathbf{F} - \mathbf{F} \cdot \operatorname{curl} \mathbf{G}$
9. $\operatorname{div} \operatorname{curl} \mathbf{F} = 0$
10. $\operatorname{curl}(f\mathbf{F}) = f\operatorname{curl} \mathbf{F} + \nabla f \times \mathbf{F}$
11. $\operatorname{curl} \nabla f = \mathbf{0}$
12. $\nabla^2(fg) = f\nabla^2 g + g\nabla^2 f + 2(\nabla f \cdot \nabla g)$
13. $\operatorname{div}(\nabla f \times \nabla g) = 0$
14. $\operatorname{div}(f\nabla g - g\nabla f) = f\nabla^2 g - g\nabla^2 f$

Here,

$$\nabla^2 f = \nabla \cdot \nabla f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

is the *Laplacian* of f .

5.1 Introduction

Key Points in this Section.

Double Integrals The volume of the region above R and under the graph of a nonnegative function f is called the **(double) integral** of f over R and is denoted by

$$\iint_R f(x, y) dA, \quad \text{or} \quad \iint_R f(x, y) dx dy.$$

Double and Iterated Integrals

$$\iint_R f(x, y) dA = \int_a^b \left[\int_c^d f(x, y) dy \right] dx. \quad (1)$$

If we use cutting planes perpendicular to the y axis, we obtain

$$\iint_R f(x, y) dA = \int_c^d \left[\int_a^b f(x, y) dx \right] dy. \quad (2)$$

The expression on the right of formula (2) is the iterated integral obtained by integrating with respect to x and then integrating the result with respect to y .

5.2 The Double Integral over a Rectangle

Key Points in this Section.

THEOREM 1 Any continuous function defined on a closed rectangle R is integrable.

THEOREM 2: Integrability of Bounded Functions Let $f: R \rightarrow \mathbb{R}$ be a bounded real-valued function on the rectangle R , and suppose that the set of points where f is discontinuous lies on a finite union of graphs of continuous functions. Then f is integrable over R .

THEOREM 3: Fubini's Theorem Let f be a continuous function with a rectangular domain $R = [a, b] \times [c, d]$. Then

$$\int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy = \iint_R f(x, y) dA. \quad (4)$$

5.3 The Double Integral Over More General Regions

Key Points in this Section.

3. For a y -simple region

$$\iint_D f dA = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y) dy dx$$

and for an x -simple region

$$\iint_D f dA = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx dy.$$

DEFINITION: Integral over an Elementary Region If D is an elementary region in the plane, choose a rectangle R that contains D . Given $f: D \rightarrow \mathbb{R}$, where f is continuous (and hence bounded), define $\iint_D f(x, y) dA$, the **integral of f over the set D** as follows: Extend f to a function f^* defined on all of R by

$$f^*(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \in D \\ 0 & \text{if } (x, y) \notin D \text{ and } (x, y) \in R. \end{cases}$$

Note that f^* is bounded (because f is) and continuous except possibly on the boundary of D (see Figure 5.3.4). The boundary of D consists of graphs of continuous functions, and so f^* is integrable over R by Theorem 2, Section 5.2. Therefore, we can define

$$\iint_D f(x, y) dA = \iint_R f^*(x, y) dA.$$

THEOREM 4': Iterated Integrals for x -Simple Regions Suppose that D is the set of points (x, y) such that $y \in [c, d]$ and $\psi_1(y) \leq x \leq \psi_2(y)$. If f is continuous on D , then

$$\iint_D f(x, y) dA = \int_c^d \left[\int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx \right] dy. \quad (2)$$

5.4 Changing the Order of Integration

Key Points in this Section.

THEOREM 5: Mean Value Theorem: Double Integrals Suppose $f: D \rightarrow \mathbb{R}$ is continuous and D is an elementary region. Then for some point (x_0, y_0) in D we have

$$\iint_D f(x, y) dA = f(x_0, y_0)A(D),$$

where $A(D)$ denotes the area of D .

5.5 The Triple Integral

Key Points in this Section.

3. **Elementary Regions.** An example of an *elementary region* W in $\mathbb{R}^3$ is one defined by inequalities $a \leq x \leq b$, $\phi_1(x) \leq y \leq \phi_2(x)$ (an elementary region in the plane) and $\gamma_1(x, y) \leq z \leq \gamma_2(x, y)$.

Triple Integrals by Iterated Integration Suppose that W is an elementary region described by bounding z between two functions of x and y . Then either

$$\iiint_W f(x, y, z) dx dy dz = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} \int_{\gamma_1(x, y)}^{\gamma_2(x, y)} f(x, y, z) dz dy dx$$

[see Figure 5.5.2 (left)] or

$$\iiint_W f(x, y, z) dx dy dz = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} \int_{\gamma_1(x, y)}^{\gamma_2(x, y)} f(x, y, z) dz dx dy$$

[see Figure 5.5.2 (right)].

6.1 The Geometry of Maps from $\mathbb{R}^2$ to $\mathbb{R}^2$

Key Points in this Section.

THEOREM 1 Let A be a 2×2 matrix with $\det A \neq 0$ and let T be the linear mapping of $\mathbb{R}^2$ to $\mathbb{R}^2$ given by $T(\mathbf{x}) = A\mathbf{x}$ (matrix multiplication). Then T transforms parallelograms into parallelograms and vertices into vertices. Moreover, if $T(D^*)$ is a parallelogram, D^* must be a parallelogram.

DEFINITION A mapping T is *one-to-one* on D^* if for (u, v) and $(u', v') \in D^*$, $T(u, v) = T(u', v')$ implies that $u = u'$ and $v = v'$.

DEFINITION The mapping T is *onto* D if for every point $(x, y) \in D$ there exists at least one point (u, v) in the domain of T such that $T(u, v) = (x, y)$.

For linear mappings T of $\mathbb{R}^n$ to $\mathbb{R}^n$ it turns out that one-to-one and onto are equivalent notions.

One-to-One and Onto Mappings A mapping $T: D^* \rightarrow D$ is **one-to-one** when it maps distinct points to distinct points. It is **onto** when the image of D^* under T is all of D .

A *linear* transformation of $\mathbb{R}^n$ to $\mathbb{R}^n$ given by multiplication by a matrix A is one-to-one and onto when and only when $\det A \neq 0$.

6.2 The Change of Variables Theorem

Key Points in this Section.

DEFINITION: Jacobian Determinant Let $T: D^* \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a C^1 transformation given by $x = x(u, v)$ and $y = y(u, v)$. The **Jacobian determinant** of T , written $\partial(x, y)/\partial(u, v)$, is the determinant of the derivative matrix $\mathbf{D}T(u, v)$ of T :

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}.$$

THEOREM 2: Change of Variables: Double Integrals Let D and D^* be elementary regions in the plane and let $T: D^* \rightarrow D$ be of class C^1 ; suppose that T is one-to-one on D^* . Furthermore, suppose that $D = T(D^*)$. Then for any integrable function $f: D \rightarrow \mathbb{R}$, we have

$$\iint_D f(x, y) dx dy = \iint_{D^*} f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv. \quad (6)$$

Change of Variables---Polar Coordinates

$$\iint_D f(x, y) dx dy = \iint_{D^*} f(r \cos \theta, r \sin \theta) r dr d\theta \quad (7)$$

Change of Variables---Cylindrical Coordinates

$$\iiint_W f(x, y, z) dx dy dz = \iiint_{W^*} f(r \cos \theta, r \sin \theta, z) r dr d\theta dz. \quad (9)$$

Change of Variables---Spherical Coordinates

$$\begin{aligned} \iiint_W f(x, y, z) dx dy dz \\ = \iiint_{W^*} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\theta d\phi. \end{aligned} \tag{10}$$

7.1 The Path Integral

Key Points in this Section.

DEFINITION: Path Integrals The *path integral*, or the *integral of* $f(x, y, z)$ *along the path* $\mathbf{c}$, is defined when $\mathbf{c}: I = [a, b] \rightarrow \mathbb{R}^3$ is of class C^1 and when the composite function $t \mapsto f(x(t), y(t), z(t))$ is continuous on I . We define this integral by the equation

$$\int_{\mathbf{c}} f ds = \int_a^b f(x(t), y(t), z(t)) \|\mathbf{c}'(t)\| dt.$$

Sometimes $\int_{\mathbf{c}} f ds$ is denoted

$$\int_{\mathbf{c}} f(x, y, z) ds$$

or

$$\int_a^b f(\mathbf{c}(t)) \|\mathbf{c}'(t)\| dt.$$

If $\mathbf{c}(t)$ is only piecewise C^1 or $f(\mathbf{c}(t))$ is piecewise continuous, we define $\int_{\mathbf{c}} f ds$ by breaking $[a, b]$ into pieces over which $f(\mathbf{c}(t)) \|\mathbf{c}'(t)\|$ is continuous, and summing the integrals over the pieces.

7.2 Line Integrals

Key Points in this Section.

DEFINITION: Line Integrals Let $\mathbf{F}$ be a vector field on $\mathbb{R}^3$ that is continuous on the C^1 path $\mathbf{c}: [a, b] \rightarrow \mathbb{R}^3$. We define $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$, the *line integral* of $\mathbf{F}$ along $\mathbf{c}$, by the formula

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t) dt;$$

that is, we integrate the dot product of $\mathbf{F}$ with $\mathbf{c}'$ over the interval $[a, b]$.

As is the case with scalar functions, we can also define $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$ if $\mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)$ is only piecewise continuous.

DEFINITION Let $h: I \rightarrow I_1$ be a C^1 real-valued function that is a one-to-one map of an interval $I = [a, b]$ onto another interval $I_1 = [a_1, b_1]$. Let $\mathbf{c}: I_1 \rightarrow \mathbb{R}^3$ be a piecewise C^1 path. Then we call the composition

$$\mathbf{p} = \mathbf{c} \circ h: I \rightarrow \mathbb{R}^3$$

a *reparametrization* of $\mathbf{c}$.

THEOREM 1: Change of Parametrization for Line Integrals

Let $\mathbf{F}$ be a vector field continuous on the C^1 path $\mathbf{c}: [a_1, b_1] \rightarrow \mathbb{R}^3$, and let $\mathbf{p}: [a, b] \rightarrow \mathbb{R}^3$ be a reparametrization of $\mathbf{c}$. If $\mathbf{p}$ is orientation-preserving, then

$$\int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s},$$

and if $\mathbf{p}$ is orientation-reversing, then

$$\int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = - \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}.$$

THEOREM 2: Change of Parametrization for Path Integrals

Let $\mathbf{c}$ be piecewise C^1 , let f be a continuous (real-valued) function on the image of $\mathbf{c}$, and let $\mathbf{p}$ be any reparametrization of $\mathbf{c}$. Then

$$\int_{\mathbf{c}} f(x, y, z) ds = \int_{\mathbf{p}} f(x, y, z) ds. \quad (2)$$

THEOREM 3: Line Integrals of Gradient Vector Fields Suppose that $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ is of class C^1 and that $\mathbf{c}: [a, b] \rightarrow \mathbb{R}^3$ is a piecewise C^1 path. Then

$$\int_{\mathbf{c}} \nabla f \cdot d\mathbf{s} = f(\mathbf{c}(b)) - f(\mathbf{c}(a)).$$

DEFINITION: Simple Closed Curves By a *simple closed curve* we mean the image of a piecewise C^1 map $\mathbf{c}: [a, b] \rightarrow \mathbb{R}^3$ that is one-to-one on $[a, b)$ and satisfies $\mathbf{c}(a) = \mathbf{c}(b)$ (Figure 7.2.8). If $\mathbf{c}$ satisfies the condition $\mathbf{c}(a) = \mathbf{c}(b)$, but is not necessarily one-to-one on $[a, b)$, we call its image a *closed curve*. Simple closed curves have two orientations, corresponding to the two possible directions of motion along the curve (Figure 7.2.9).

Line Integrals Over Oriented Simple Curves and Simple Closed Curves C :

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} \quad \text{and} \quad \int_C f ds = \int_{\mathbf{c}} f ds, \quad (3)$$

where $\mathbf{c}$ is any *orientation-preserving* parametrization of C .

Line Integrals Over Curves with Opposite Orientations Let C^- be the same curve as C , but with the opposite orientation. Then

$$\int_{C^-} \mathbf{F} \cdot d\mathbf{s} = - \int_C \mathbf{F} \cdot d\mathbf{s}.$$

Line Integrals Over Curves Consisting of Several Components Let C be an oriented curve that is made up of several oriented component curves C_i , $i = 1, \dots, k$, as in Figure 7.2.11. Then we shall write $C = C_1 + C_2 + \dots + C_k$. Because we can parametrize C by parametrizing the pieces $C_1, \dots, C_k$ separately, one can prove that

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_{C_1} \mathbf{F} \cdot d\mathbf{s} + \int_{C_2} \mathbf{F} \cdot d\mathbf{s} + \dots + \int_{C_k} \mathbf{F} \cdot d\mathbf{s}. \quad (4)$$

7.3 Parametrized Surfaces

Key Points in this Section.

DEFINITION: Parametrized Surfaces A *parametrized surface* is a function $\Phi: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$, where D is some domain in $\mathbb{R}^2$. The *surface* S corresponding to the function Φ is its image: $S = \Phi(D)$. We can write

$$\Phi(u, v) = (x(u, v), y(u, v), z(u, v)).$$

If Φ is differentiable or is of class C^1 [which is the same as saying that $x(u, v)$, $y(u, v)$, and $z(u, v)$ are differentiable or C^1 functions of (u, v)], we call S a *differentiable or a C^1 surface*.

4. Tangent vectors to the surface are given by

$$\mathbf{T}_u = \frac{\partial \Phi}{\partial u} = \frac{\partial x}{\partial u} \mathbf{i} + \frac{\partial y}{\partial u} \mathbf{j} + \frac{\partial z}{\partial u} \mathbf{k}.$$

and

$$\mathbf{T}_v = \frac{\partial \Phi}{\partial v} = \frac{\partial x}{\partial v} \mathbf{i} + \frac{\partial y}{\partial v} \mathbf{j} + \frac{\partial z}{\partial v} \mathbf{k}.$$

with a normal vector being given by

$$\mathbf{n} = \mathbf{T}_u \times \mathbf{T}_v.$$

7.4 Area of a Surface

Key Points in this Section.

DEFINITION: Area of a Parametrized Surface We define the *surface area*¹⁰ $A(S)$ of a parametrized surface by

$$A(S) = \iint_D \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv \quad (1)$$

where $\|\mathbf{T}_u \times \mathbf{T}_v\|$ is the norm of $\mathbf{T}_u \times \mathbf{T}_v$. If S is a union of surfaces S_i , its area is the sum of the areas of the S_i .

6. Surface area of a graph.

$$A(S) = \iint_D \left(\sqrt{\left[\frac{\partial g}{\partial x} \right]^2 + \left[\frac{\partial g}{\partial y} \right]^2 + 1} \right) du \, dv$$

7.5 Integrals of Scalar Functions over Surfaces

Key Points in this Section.

DEFINITION: The Integral of a Scalar Function Over a Surface If $f(x, y, z)$ is a real-valued continuous function defined on a parametrized surface S , we define the *integral of f over S* to be

$$\iint_S f(x, y, z) dS = \iint_S f dS = \iint_D f(\Phi(u, v)) \|\mathbf{T}_u \times \mathbf{T}_v\| du dv. \quad (1)$$

7.6 Surface Integrals of Vector Functions

Key Points in this Section.

DEFINITION: The Surface Integral of Vector Fields Let $\mathbf{F}$ be a vector field defined on S , the image of a parametrized surface Φ . The *surface integral of $\mathbf{F}$ over Φ* , denoted by

$$\iint_{\Phi} \mathbf{F} \cdot d\mathbf{S},$$

is defined by (see Figure 7.6.1))

$$\iint_{\Phi} \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) du dv.$$

DEFINITION: Oriented Surfaces An *oriented surface* is a two-sided surface with one side specified as the *outside* or *positive side*; we call the other side the *inside* or *negative side*.¹³ At each point $(x, y, z) \in S$ there are two unit normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, where $\mathbf{n}_1 = -\mathbf{n}_2$ (see Figure 7.6.2). Each of these two normals can be associated with one side of the surface. Thus, to specify a side of a surface S , at each point we choose a unit normal vector $\mathbf{n}$ that points away from the positive side of S at that point.

The Surface Integral of a Vector Field Over a Graph S

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_D \mathbf{F} \cdot (\mathbf{T}_x \times \mathbf{T}_y) dx dy \\ &= \iint_D \left[F_1 \left(-\frac{\partial g}{\partial x} \right) + F_2 \left(-\frac{\partial g}{\partial y} \right) + F_3 \right] dx dy. \end{aligned} \quad (4)$$

8.1 Green's Theorem

Key Points in this Section.

THEOREM 1: Green's Theorem Let D be a simple region and let C be its boundary. Suppose $P: D \rightarrow \mathbb{R}$ and $Q: D \rightarrow \mathbb{R}$ are of class C^1 . Then

$$\int_{C^+} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

THEOREM 2: Area of a Region If C is a simple closed curve that bounds a region to which Green's theorem applies, then the area of the region D bounded by $C = \partial D$ is

$$A = \frac{1}{2} \int_{\partial D} x dy - y dx.$$

THEOREM 3: Vector Form of Green's Theorem Let $D \subset \mathbb{R}^2$ be a region to which Green's theorem applies, let ∂D be its boundary (oriented counterclockwise), and let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ be a C^1 vector field on D . Then

$$\int_{\partial D} \mathbf{F} \cdot d\mathbf{s} = \iint_D (\text{curl } \mathbf{F}) \cdot \mathbf{k} dA = \iint_D (\nabla \times \mathbf{F}) \cdot \mathbf{k} dA$$

(see Figure 8.1.7).

THEOREM 4: Divergence Theorem in the Plane Let $D \subset \mathbb{R}^2$ be a region to which Green's theorem applies and let ∂D be its boundary. Let $\mathbf{n}$ denote the outward unit normal to ∂D . If $\mathbf{c}: [a, b] \rightarrow \mathbb{R}^2, t \mapsto \mathbf{c}(t) = (x(t), y(t))$ is a positively oriented parametrization of ∂D , $\mathbf{n}$ is given by

$$\mathbf{n} = \frac{(y'(t), -x'(t))}{\sqrt{[x'(t)]^2 + [y'(t)]^2}}$$

(see Figure 8.1.9). Let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ be a C^1 vector field on D . Then

$$\int_{\partial D} \mathbf{F} \cdot \mathbf{n} ds = \iint_D \text{div } \mathbf{F} dA.$$

8.2 Stokes' Theorem

Key Points in this Section.

THEOREM 5: Stokes' Theorem for Graphs Let S be the oriented surface defined by a C^2 function $z = f(x, y)$, where $(x, y) \in D$, a region to which Green's theorem applies, and let $\mathbf{F}$ be a C^1 vector field on S . Then if ∂S denotes the oriented boundary curve of S as just defined, we have

$$\iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}.$$

8.4 Gauss' Theorem

Key Points in this Section.

THEOREM 9: Gauss' Divergence Theorem Let W be a symmetric elementary region in space. Denote by ∂W the oriented closed surface that bounds W . Let $\mathbf{F}$ be a smooth vector field defined on W . Then

$$\iiint_W (\nabla \cdot \mathbf{F}) dV = \iint_{\partial W} \mathbf{F} \cdot d\mathbf{S}$$

or, alternatively,

$$\iiint_W (\operatorname{div} \mathbf{F}) dV = \iint_{\partial W} (\mathbf{F} \cdot \mathbf{n}) dS.$$